

Lab 2: Introduction to Packet Tracer

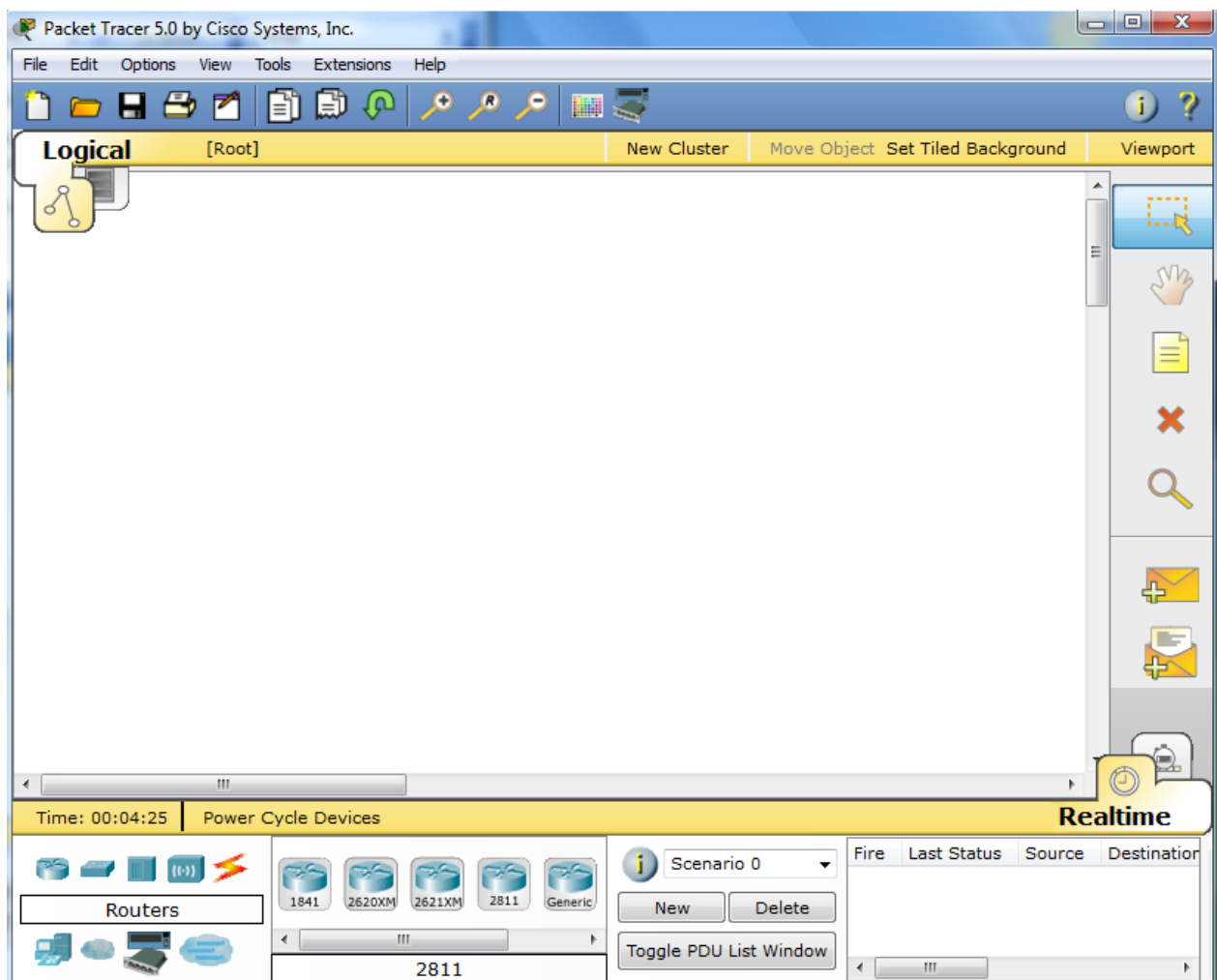
Objectives

The purpose of this lab is to become familiar with building topologies in Packet Tracer.

Introduction

Packet Tracer is a protocol simulator developed by Dennis Frezzo and his team at Cisco Systems. Packet Tracer (PT) is a powerful and dynamic tool that displays the various protocols used in networking, in either Real Time or Simulation mode. This includes layer 2 protocols such as Ethernet and PPP, layer 3 protocols such as IP, ICMP, and ARP, and layer 4 protocols such as TCP and UDP. Routing protocols can also be traced.

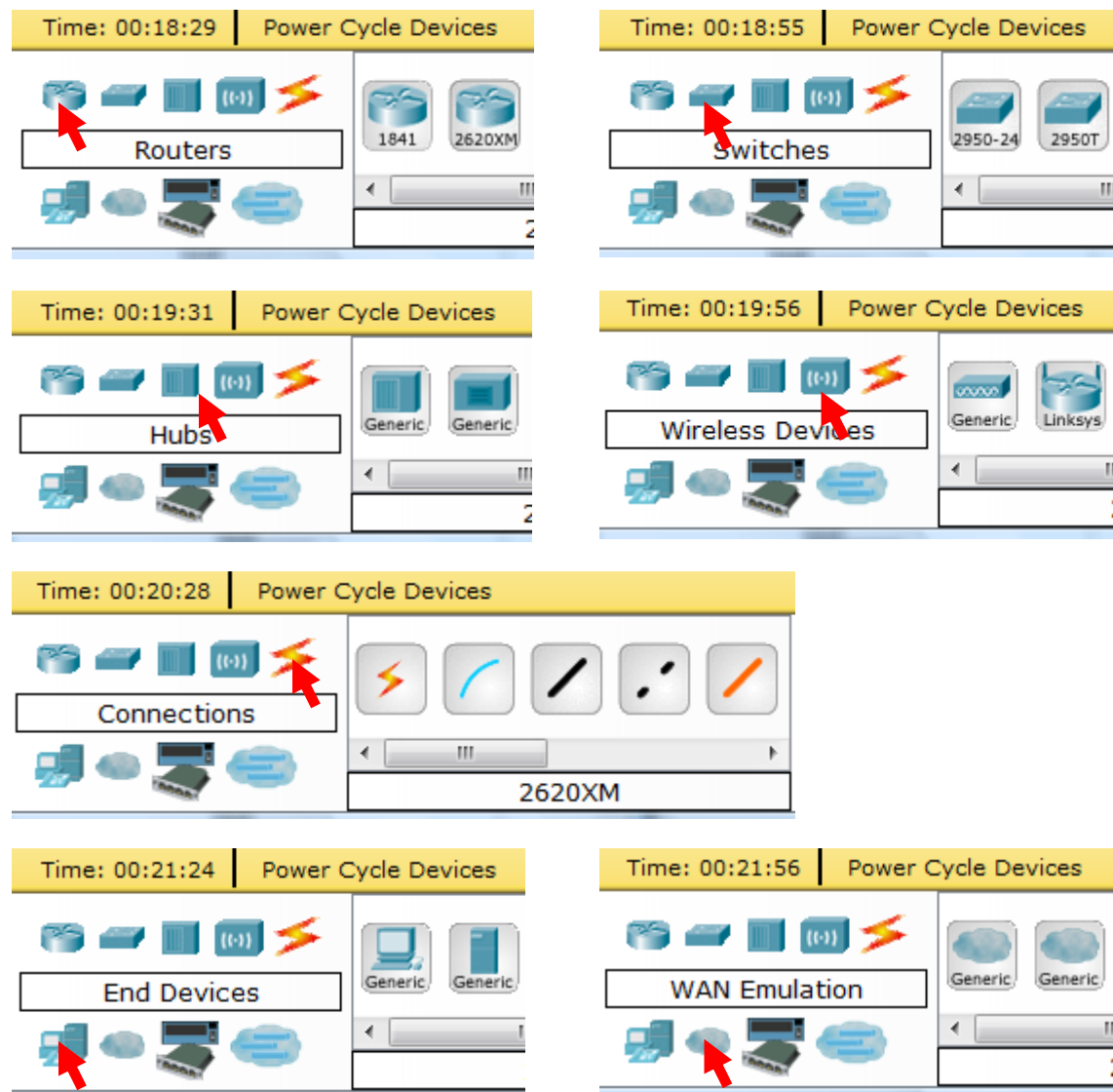
Step 1: Start Packet Tracer

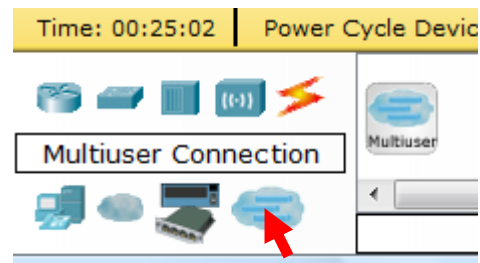
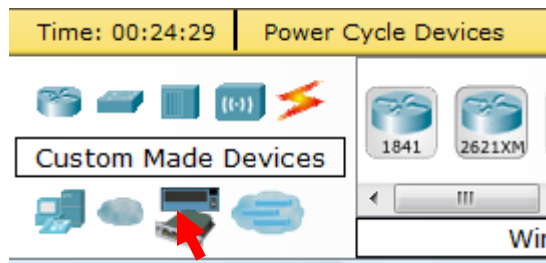


Step 2: Choosing Devices and Connections

We will begin building our network topology by selecting devices and the media in which to connect them. Several types of devices and network connections can be used. For this lab we will keep it simple by using **End Devices**, **Switches**, **Hubs**, and **Connections**.

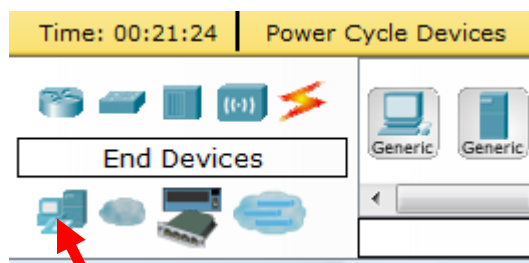
Single click on each group of devices and connections to display the various choices. The devices you see may differ slightly.



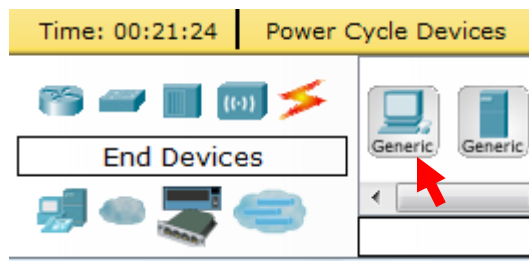


Step 3: Building the Topology – Adding Hosts

Single click on the **End Devices**.



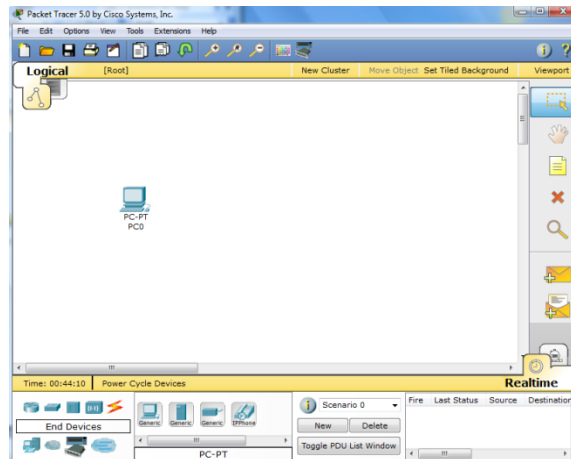
Single click on the **Generic** host.



Move the cursor into topology area. You will notice it turns into a plus “+” sign.

+

Single click in the topology area and it copies the device.



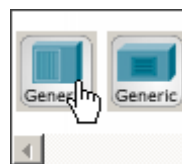
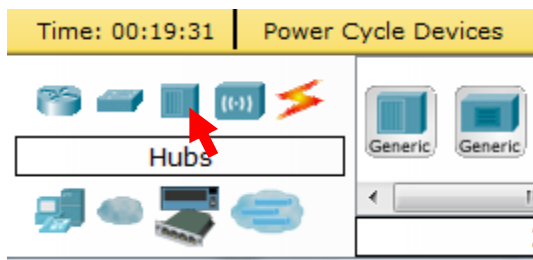
Add three more hosts.



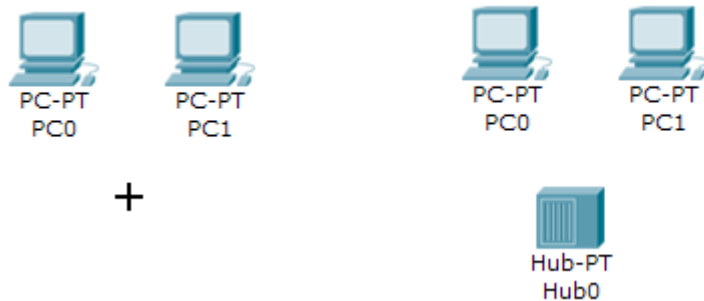
Step 4: Building the Topology – Connecting the Hosts to Hubs and Switches

Adding a Hub

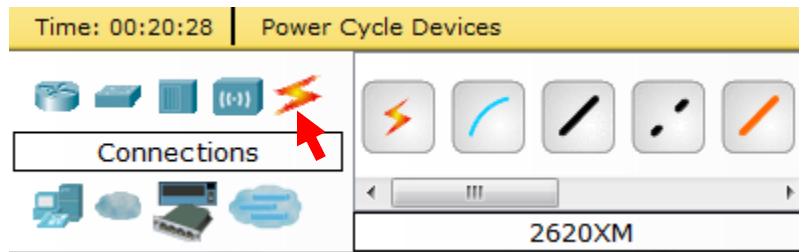
Select a hub, by clicking once on **Hubs** and once on a **Generic** hub.



Add the hub by moving the plus sign “+” below PC0 and PC1 and click once.



Connect PC0 to Hub0 by first choosing **Connections**.



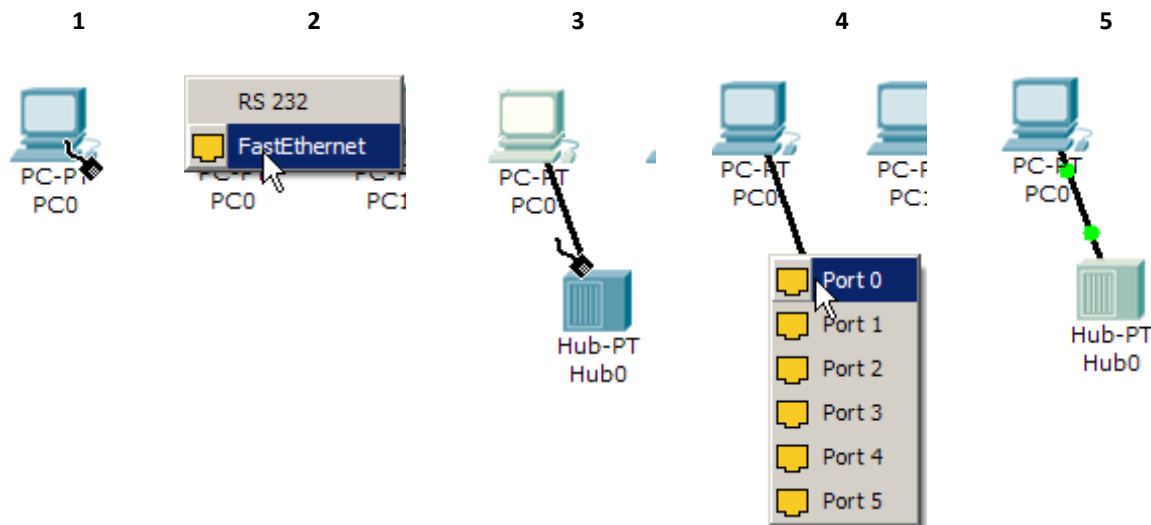
Click once on the **Copper Straight-through** cable.



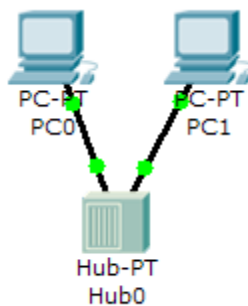
Perform the following steps to connect **PC0** to **Hub0**:

1. Click once on **PC0**
2. Choose **FastEthernet**

3. Drag the cursor to **Hub0**
4. Click once on **Hub0** and choose **Port 0**
5. Notice the green link lights on both the **PC0** Ethernet NIC and the **Hub0** Port 0 showing that the link is active.

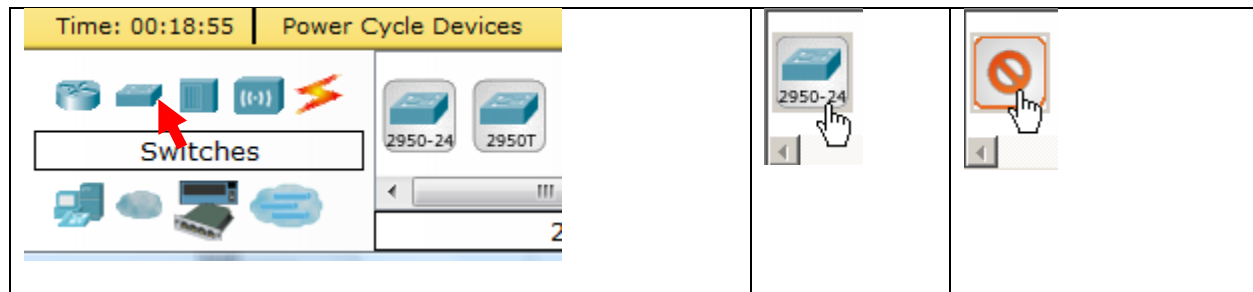


Repeat the steps above for **PC1** connecting it to **Port 1** on **Hub0**. (The actual hub port you choose does not matter.)

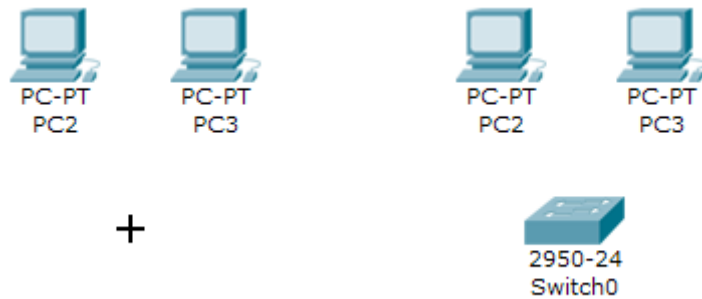


Adding a Switch

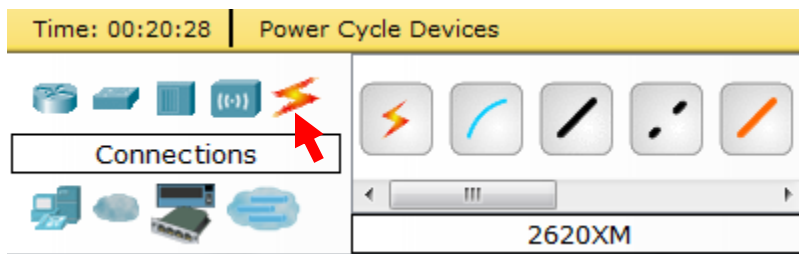
Select a switch, by clicking once on **Switches** and once on a **2950-24** switch.



Add the switch by moving the plus sign “+” below PC2 and PC3 and click once.



Connect PC2 to Hub0 by first choosing **Connections**.

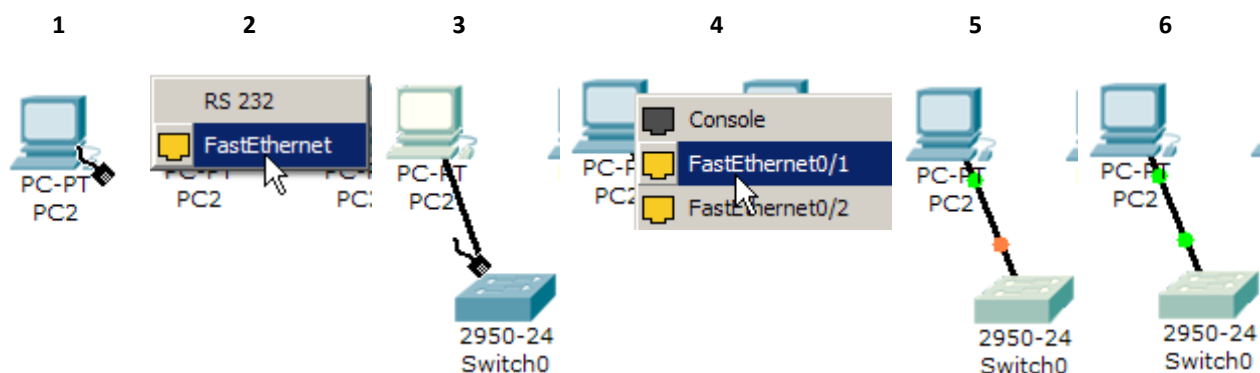


Click once on the **Copper Straight-through** cable.



Perform the following steps to connect **PC2** to **Switch0**:

1. Click once on **PC2**
2. Choose **FastEthernet**
3. Drag the cursor to **Switch0**
4. Click once on **Switch0** and choose **FastEthernet0/1**
5. Notice the green link lights on **PC2** Ethernet NIC and amber light **Switch0 FastEthernet0/1 port**. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.
6. After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now forwarded out the switch port.



Repeat the steps above for **PC3** connecting it to **Port 3** on **Switch0** on port **FastEthernet0/2**. (The actual switch port you choose does not matter.)

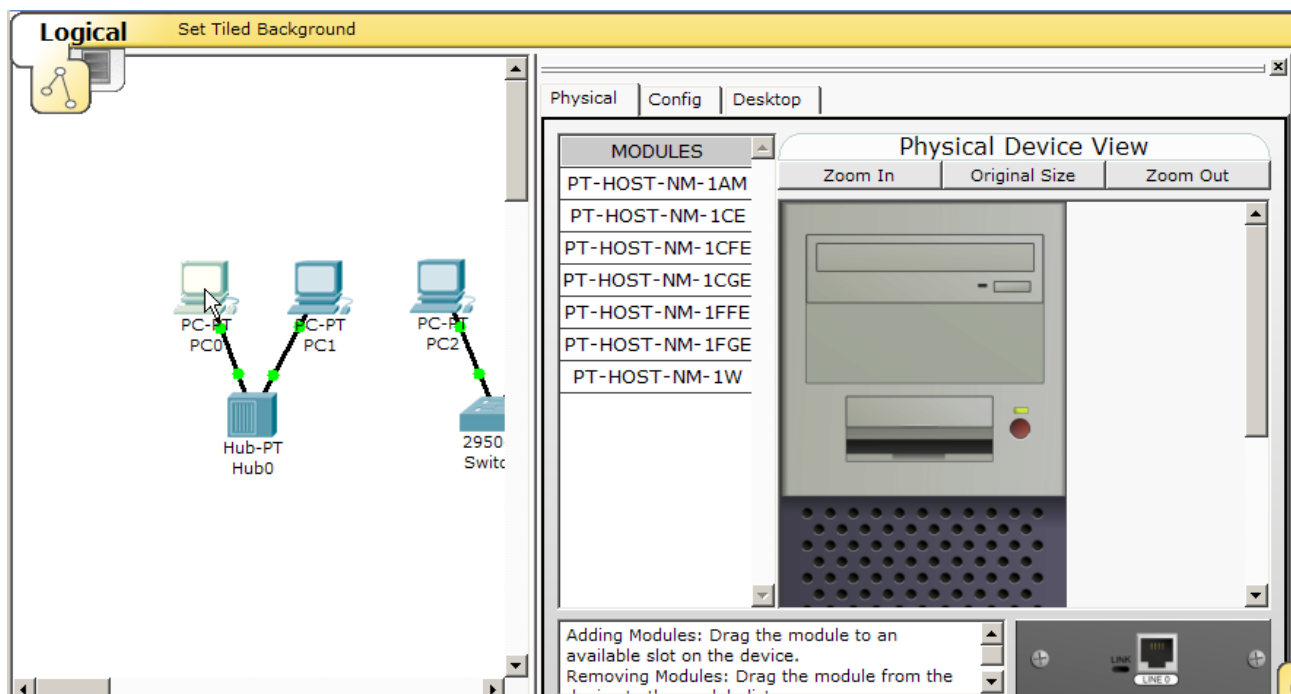


Move the cursor over the link light to view the port number. **Fa** means FastEthernet, 100 Mbps Ethernet.

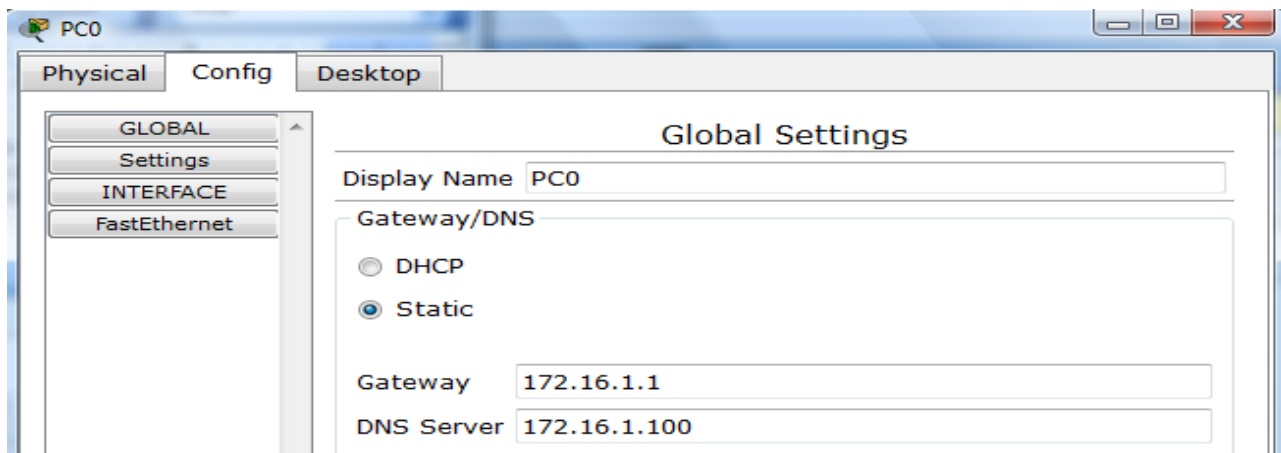
Step 5: Configuring IP Addresses and Subnet Masks on the Hosts

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

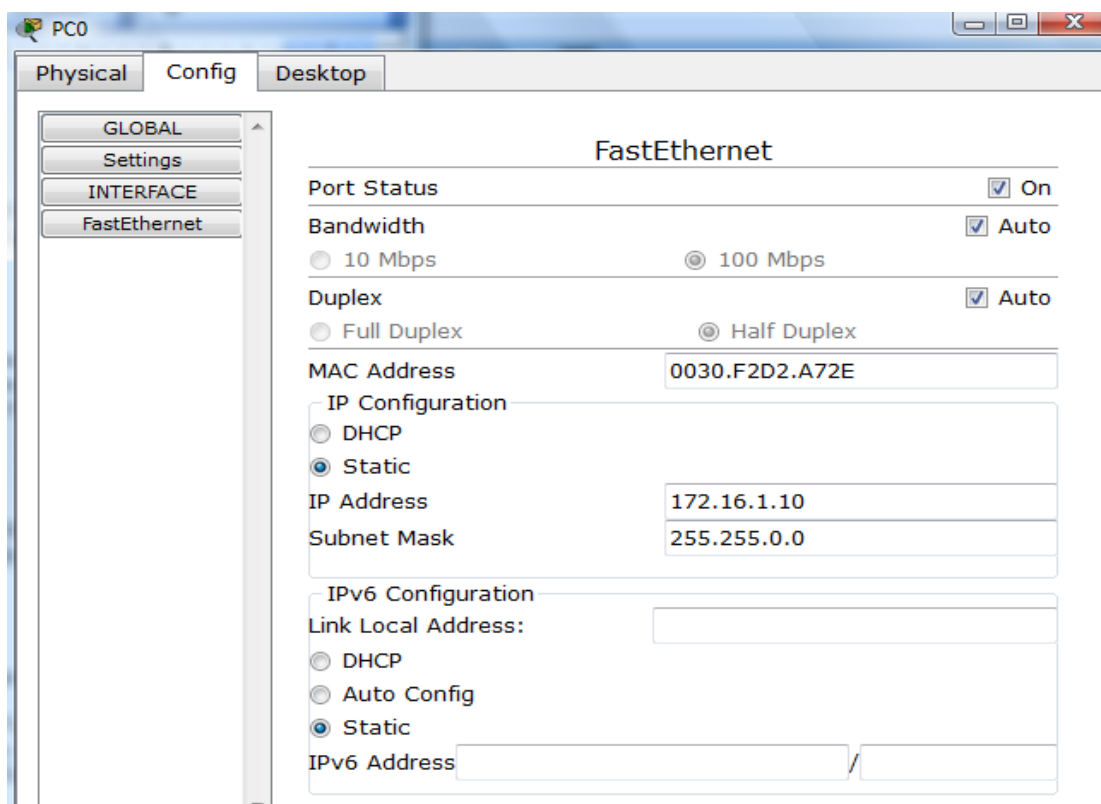
Click once on PC0.



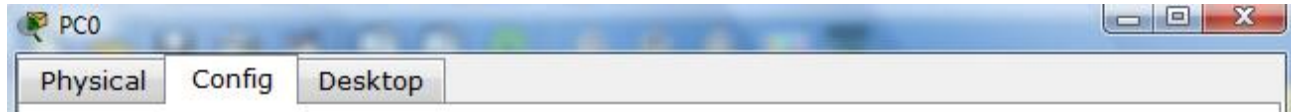
Choose the **Config** tab and click on **Settings**. It is here that you can change the name of PC0. It is also here where you would enter a **Gateway** IP Address, also known as the default gateway and the **DNS Server** IP Address. We will discuss this later, but this would be the IP address of the local router. If you want, you can enter the Gateway IP Address 172.16.1.1 and DNS Server IP Address 172.16.1.100, although it will not be used in this lab.



Click on **Interface** and then **FastEthernet**. Although we have not yet discussed IP Addresses, add the IP Address to 172.16.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.0.0. We will discuss this later.



To close this dialog box, click the “X” in the upper right.

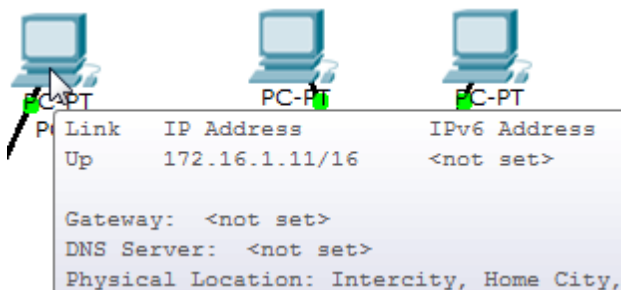


Repeat these steps for the other hosts. Use the information below for IP Addresses and Subnet Masks.

<u>Host</u>	<u>IP Address</u>	<u>Subnet Mask</u>
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

Verify the information

To verify the information that you entered, move the Select tool (arrow) over each host.



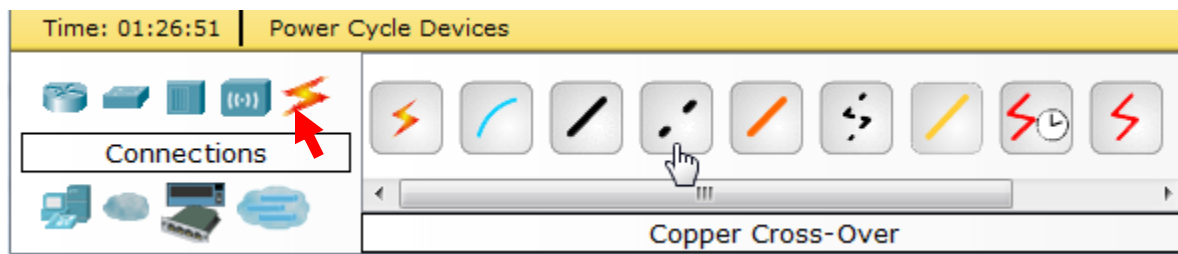
Deleting a Device or Link

To delete a device or link, choose the **Delete** tool and click on the item you wish to delete.

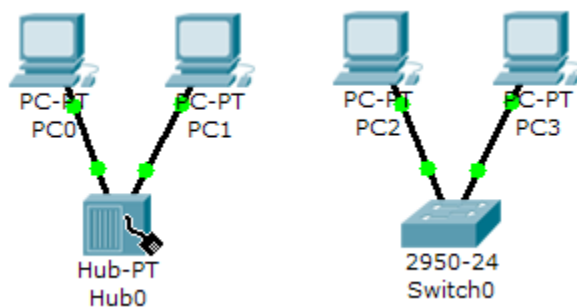


Step 6: Connecting Hub0 to Switch0

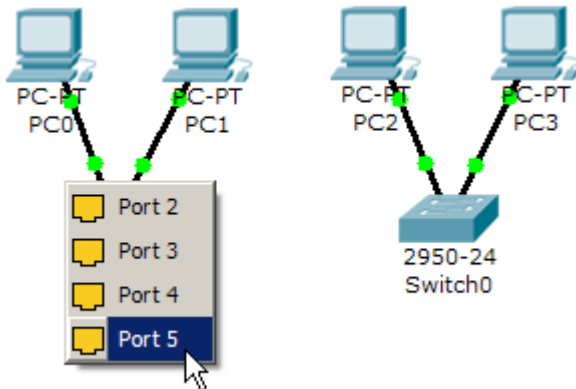
To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable. Click once the **Cross-over** Cable from the **Connections** options.



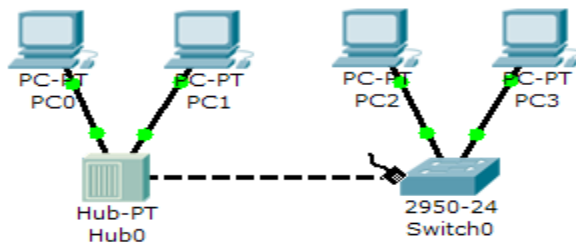
Move the Connections cursor over **Hub0** and click once.



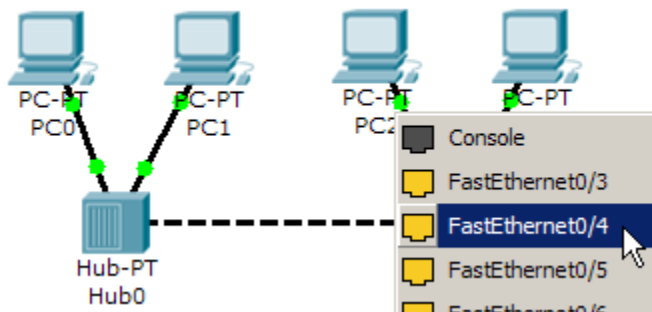
Select **Port 5** (actual port does not matter).



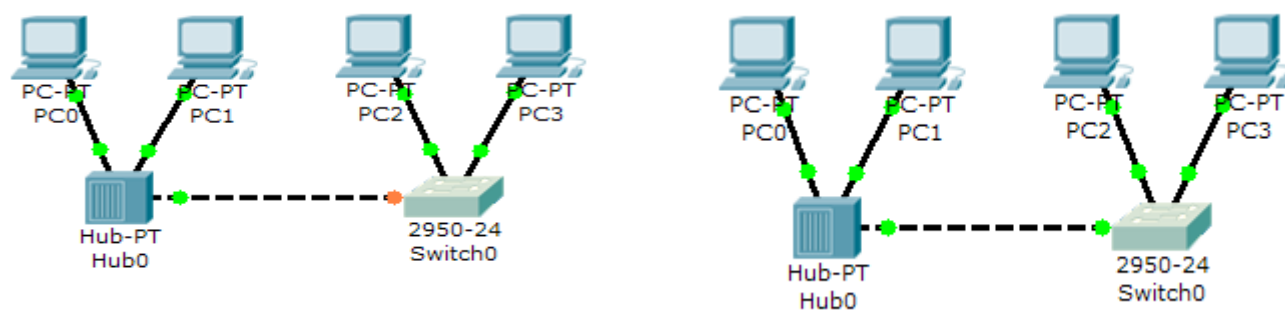
Move the Connections cursor to **Switch0**.



Click once on **Switch0** and choose **FastEthernet0/4** (actual port does not matter).



The link light for switch port **FastEthernet0/4** will begin as amber and eventually change to green.



Lab Tasks**Post Lab Task 1:**

Write the advantages and disadvantages of different network topologies (Mesh, Star, Bus and ring)?

i. Mesh:

ii. Star:

iii. Bus

iv. Ring

Post Lab Task 2:

Implement the following topologies in packet tracer and attach screen shots ((Mesh, Star, and ring)?